Grade 9 – Mathematics

Third Term Model Paper 3

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Name / Index No.: ..... **Time: 3 hours**

- This paper consists of two parts: Part I and Part II.
- Part I: Answer **all** questions on this paper itself. Part A has 15 questions (2 marks each). Part B has 2 questions (10 marks each).
- Part II: Answer **five** questions only, on separate sheets. Each question carries 10 marks.
- Show all relevant steps of working. Write units where necessary. Use π = 22/7 where needed.

For examiner's use only	
PART I – A	/ 30
PART I – B	/ 20
PART II	/ 50
Total	**/ 100**

PART I

Part A – Answer all questions in the space provided. (2 marks each)

1. Write the binary number 110110_2 as a decimal number (in base ten). (2)

Answer: .....

2. The nth term of a number pattern is $n^2 + 1$. Write its 4th and 5th terms. (2)

Answer: .....

3. Simplify: $1\frac{2}{3} + 2\frac{3}{4} - 1\frac{1}{2}$ (2)

Answer: .....

4. The price of 4 books is Rs 960. Find the price of 7 such books. (2)

Answer: .....

5. Factorise: $4x^2 - 25$ (2)

Answer: .....

6. Simplify: $x^{-2} \times x^5 \times y^0$ (2)

Answer: .....

7. Round off 0.04572 to two significant figures. (2)

Answer: .....

8. Solve: $x/3 + 2 = 7$ (2)

Answer: .....

9. In the figure, A, B, C and D are points on a straight line and AB = CD. Show that AC = BD, and name the axiom used. (2)



Answer:

.....

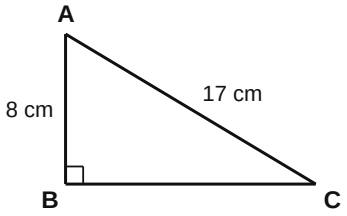
10. The angles of a triangle are in the ratio 2 : 3 : 4. Find the largest angle. (2)

Answer: .....

11. Find the perimeter of a semicircular lamina of radius 14 cm. ($\pi = 22/7$) (2)

Answer: .....

12. Find the length of BC in the right-angled triangle ABC. (2)



Answer: .....

13. $n(A) = 12$, $n(B) = 9$ and $n(A \cap B) = 4$. Find $n(A \cup B)$. (2)

Answer: .....

14. Find the magnitude of an interior angle of a regular decagon (10 sides). (2)

Answer: .....

15. How many 750 ml bottles can be completely filled from 6 litres of juice? (2)

Answer: .....

Part B – Answer both questions on this paper. (10 marks each)

1.

The figures below are made with matchsticks. The number of matchsticks in each figure forms a number pattern.



(i) Write the number of matchsticks in each of the first three figures. (2 marks)

(ii) Write the general term (n^{th} term) of the pattern. (2 marks)

(iii) How many matchsticks are needed for the 25th figure? (2 marks)

(iv) Which figure needs 61 matchsticks? (2 marks)

(v) Can a figure in this pattern be made with exactly 100 matchsticks? Give a reason. (2 marks)

2.

The monthly expenses of a family, which total Rs 60 000, are given below.

Item	Food	Education	Transport	Other
Amount (Rs)	24 000	12 000	9 000	15 000

(i) Find the angle of the sector representing each item in a pie chart. (4 marks)

(ii) Draw a pie chart (radius 3 cm) to represent this information. (3 marks)

(iii) What percentage of the total expenses is spent on transport? (3 marks)

PART II

Answer five questions only. (10 marks each)

1.

(a) A trader buys 200 mangoes at Rs 60 each. 20 of them get spoiled. He sells the rest at Rs 80 each.

(i) Find the total cost and the total income. (2 marks)

(ii) Find the profit as a percentage of the cost. (3 marks)

(b) The distance from the Earth to the Sun is about 150 000 000 km.

(i) Write this distance in scientific notation. (2 marks)

(ii) Light travels 3×10^5 km per second. Find the time taken for light from the Sun to reach the Earth, in seconds. (3 marks)

2.

(a) Simplify: (i) $(3a^2b)^2$ (ii) $2^5 \times 2^{-3} \div 2^{-1}$ (iii) $(x^3y^{-2})^0 \times 5$ (5 marks)

(b) Find the value of x: (i) $2^x = 32$ (ii) $3^{x+1} = 81$ (3 marks)

(c) Find the value of $5.2 \times 10^3 + 4.8 \times 10^3$ and write the answer in scientific notation. (2 marks)

3.

(a) Solve the inequality $2(x - 1) > x + 3$, represent the solution on a number line, and write the smallest integer that satisfies it. (3 marks)

(b) An incomplete table of values for $y = -2x + 3$ is given below.

x	–1	0	1	2	3
y	5	…	1	…	–3

(i) Complete the table and draw the graph on graph paper. (4 marks)

(ii) Write the gradient and the intercept. (2 marks)

(iii) Show that the point (4, –5) lies on this line. (1 marks)

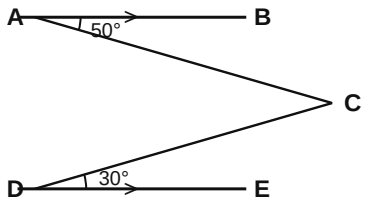
4.

(a) Solve the simultaneous equations: $3x + 2y = 16$, $x - y = 2$ (4 marks)

(b) Solve: $3(2x - 1)/5 = 3$ (2 marks)

(c) The formula $C = 5(F - 32)/9$ converts a temperature in °F to °C. (i) Find C when $F = 95$. (ii) Make F the subject of the formula. (4 marks)

5.



(a) In the figure, AB ∥ DE, BÂC = 50° and CD̂E = 30°. Find AĈD, giving reasons. (4 marks)
(Hint: draw a line through C parallel to AB.)

(b) In triangle PQR, PQ = PR and PQ̂R = 65°. Find QP̂R. (2 marks)

(c) Each interior angle of a regular polygon is five times each exterior angle. Find the exterior angle and the number of sides. (4 marks)

6.

Use only a straight edge with a cm/mm scale and a pair of compasses for the following constructions. Show all construction lines.

(i) Draw a straight line segment PQ of length 8 cm. (1 marks)

(ii) Construct an angle of 90° at P. (3 marks)

(iii) Mark R on the perpendicular arm such that PR = 6 cm, and complete the triangle PQR. (1 marks)

(iv) Measure and write the length of QR. Verify your answer using Pythagoras' relation. (2 marks)

(v) Construct the perpendicular bisector of QR, name the point where it meets QR as M, and measure PM. (3 marks)

7.

(a) A field is in the shape of a trapezium. Its parallel sides are 30 m and 20 m, and the perpendicular distance between them is 16 m.

(i) Find the area of the field. (2 marks)

(ii) Find the cost of planting grass on the whole field at Rs 125 per square metre. (2 marks)

(b) A fair die is rolled once.

(i) Write the sample space. (1 marks)

(ii) Find the probability of getting a number greater than 4. (2 marks)

(iii) Find the probability of getting a factor of 6. (2 marks)

(iv) Find the probability of not getting a prime number. (1 marks)